Defining Reality in Dementia VR: Stakeholder Perspectives on Ecological Validity for Functional Activity Training

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Abstract

Virtual reality (VR) is increasingly used in dementia care, yet most applications focus on recreation or cognitive stimulation rather than supporting the everyday activities that matter for independent living. To understand what makes VR practice feel realistic and useful for people with dementia (PwD), we conducted semi-structured interviews with PwD, caregivers, and therapists using visual probes grounded in daily living contexts. We examine how stakeholders define realism and usefulness in VR-based support for instrumental activities of daily living (IADLs) and how these judgments relate to the concept of ecological validity. Our findings show that realistic IADL-based VR is characterized by environments and task flows that align with the cognitive and functional demands of real-world activities, while useful VR evokes behavior that meaningfully reflects everyday performance and supports rehabilitation practice. We translate these insights into design implications for at-home IADL-focused VR systems that emphasize functional fidelity, adaptability, and collaborative use, grounding real-world relevance in the lived routines and caregiving ecosystems of PwD.

CCS Concepts

• **Human-centered computing → Empirical studies in HCI**.

Keywords

Dementia, Virtual Reality, Ecological Validity, Instrumental Activities of Daily Living (IADLs), Functional Fidelity

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1 Introduction

Dementia is a neurodegenerative condition that affects more than 55 million people worldwide and progressively disrupts memory, reasoning, and the ability to manage daily life [53, 54]. Because pharmacological treatments—primarily drug-based therapies—offer only modest and temporary benefits, rehabilitation approaches have become central to supporting the well-being of people with dementia (PwD) and their caregivers [28, 75, 80, 82].

A consistent insight across rehabilitation is that PwDs maintain abilities longer when they continue participating in everyday activities that matter to them [23, 38]. Tasks such as meal preparation, grocery planning, or managing household routines—often classified in rehabilitation contexts as instrumental activities of daily living (IADLs)—are not only practical but deeply tied to identity, purpose, and quality of life [6, 34, 40, 49, 52, 55, 63, 70]. Practice is most effective when these activities resemble the real situations people encounter in daily life, yet supporting such practice can be difficult. Real-world tasks often require extensive setup, pose potential safety risks, and often exceed the time and energy available to caregivers and clinicians [75]. This raises a central question for rehabilitation and technology design: what does it mean for PwD to practice everyday activities in ways that reflect the realities of their daily lives?

Virtual reality (VR) offers a promising complement by providing safe, controlled environments where IADLs can be practiced repeatedly without the logistical or safety barriers of real-world settings [10, 21, 60]. A core motivation for using VR in rehabilitation is the intuitive belief that simulated environments can provide *real-world-like* practice. In clinical and psychological research, this idea is often discussed using the term **ecological validity**, which refers to the extent to which an activity resembles, and predicts, real-life performance. However, in both rehabilitation and HCI, ecological

validity is frequently invoked without clear definition or operationalization. Many VR systems focus on visual detail, immersion, or replications of traditional neuropsychological tests, while paying less attention to the lived routines, priorities, and challenges of PwD [3, 33, 51, 65, 74]. As a result, there remains limited understanding of what *realistic* and *useful* VR practice should entail for IADL-focused systems and how it might be grounded in the everyday contexts that shape PwD's lives.

HCI researchers have long argued that dementia technologies must look beyond cognitive decline to engage with the personhood, relationships, and everyday environments of PwD [29, 39]. This perspective has inspired VR systems for recreation [11, 20, 29, 46, 47, 64, 76], reminiscence [71], and physical engagement [18, 35]. While these systems can provide emotional or social benefits, they rarely support everyday functional activities or independence. Within the CHI community, there is growing awareness of the need for more nuanced engagement with vulnerable populations [29, 39]. Even so, progress on VR intended to directly support daily living has been constrained by cautious design practices and limited integration of rehabilitation frameworks. One of the few efforts to bridge these disciplinary boundaries is Dole and Ju's work [17], which articulates ecological validity and the VR elements needed to support it within search-and-rescue contexts. Yet, this articulation has not been extended to dementia or to VR systems designed to support rehabilitation and IADL practice, leaving open important questions about how ecological validity should be understood in everyday activity contexts.

To move beyond these limitations, we examine how PwD, caregivers, and therapists define what makes VR realistic and useful for practicing everyday activities at home. Using semi-structured interviews supported by visual probes, we explore which aspects of tasks, environments, assistance, and caregiving relationships matter when imagining VR-based support. This approach allows us to shift ecological validity away from assumptions about immersion or visual fidelity and toward a situated understanding grounded in the functional and cognitive dimensions of IADLs.

With this in mind, we address the following research questions:

- (RQ1) How do PwD, caregivers, and therapists define *realistic* and *useful* in VR for everyday living activities?
- (RQ2) How can VR be designed to achieve ecological validity by aligning with lived experiences and daily priorities?

Our study contributes to the HCI community by reframing ecological validity as a situated quality defined in practice by PwD, caregivers, and OTs. We identify opportunities for VR to function not only as a training tool but as a collaborative platform that promotes autonomy, shared understanding, and meaningful participation in IADLs. In doing so, we extend conversations about designing for vulnerable populations and show how interdisciplinary approaches can strengthen VR design.

Glossary

Ecological Validity: The extent to which an activity resembles, and predicts, real-life performance; investigations of both verisimilitude and veridicality (p. 96) [22].

Verisimilitude: The similarity of [VR] to tasks and skills required in the free and open environment (p. 93) [22].

Veridicality: The extent to which performance on a task can predict phenomena in the... *real world* (p. 93) [22].

Face Validity: The degree to which a study or particular aspect thereof appears to accomplish its intended aims, regardless of whether it actually does (p.6) [17].

Realism: The extent to which a virtual environment aligns with the real world in appearance, function, or context, encompassing both objective coherence and the user's subjective sense of believability [2, 26, 72, 79].

Immersion: The objective, technical capabilities of the hardware. It captures the extent to which the VR system can isolate users from the physical world and engage their senses [2, 72, 79].

Visual Fidelity: How accurately visual details (e.g., lighting, textures, scale, motion) replicate the real world [2, 72].

Functional Fidelity: The extent to which virtual objects and tools react to user actions in the same manner as their real-world equivalents [2, 72].

Presence: The feeling of *being there* in the virtual environment. It reflects the extent to which users feel absorbed and engaged, and whether they respond to the virtual environment as though it were real [2, 14, 79].

2 Related Work

We review prior VR work for dementia, examining how assumptions about PwD and ecological validity shape design choices and where conceptual and practical gaps remain in supporting everyday activities. For clarity, we provide a glossary of key terms (see Glossary) that we use throughout the paper.

2.1 Ecological Validity in Neuropsychology-Driven VR

Many IADL-based VR systems for dementia originate in neuropsychology and are framed as modernized versions of paper-and-pencil assessments. These assessments use highly structured tasks administered on paper and measure isolated cognitive processes [41, 73] rather than the multi-step, context-rich nature of everyday activities. These systems frequently assume that greater immersion or visual fidelity inherently increases ecological validity and, consequently, the ecological validity of assessment tasks [3, 33, 60, 65, 74]. Yet, across this literature, ecological validity is rarely defined and is often conflated with **immersion**, being treated as a catch-all for hardware-driven sensory isolation (for example, using head-mounted displays to block out the physical world) or simply adding higher **visual fidelity** (replication of real-world visual detail). This

conflation obscures important conceptual distinctions and weakens claims about real-world applicability, since immersive hardware alone does not guarantee functional relevance or alignment with everyday task demands [60, 74].

This ambiguity extends more broadly across VR research in neuropsychology, prompting recent calls to clarify the foundations of ecological validity in VR [30, 62, 74]. These discussions often converge on two widely recognized components: **verisimilitude**, the similarity between a task and its real-world demands, and **veridicality**, the extent to which performance predicts real-world outcomes [22]. Studies involving populations such as PwD often prioritize verisimilitude—how closely VR mirrors everyday task structure—over long-term predictive accuracy [74]. Although these distinctions are conceptually established, they remain rarely operationalized in VR design. Without shared definitions, researchers risk making inconsistent or overstated claims about ecological validity. Clearer conceptual grounding is necessary to evaluate whether VR tasks meaningfully reflect the lived demands of daily activities.

2.2 VR as a Rehabilitation Complement

Rehabilitation-oriented VR systems shift the focus away from cognitive assessment and toward supporting ongoing engagement in IADLs through repeated, goal-directed practice. In dementia care, this orientation foregrounds functional participation, personal routines, and the lived experience of navigating daily tasks, positioning VR as a complement to—not a replacement for—embodied, real-world activity. However, existing work often falls short of these aims, with limited attention to how IADLs align with the lived routines of PwD, minimal reporting on participants' contributions to task design, and a continued inherit assumptions from neuropsychological assessment. Oliveira et al. [51], for example, adapted the Systemic Lisbon Battery [24] but offered no justification for task selection or modifications for PwD. Although they reported gains in isolated cognitive domains, the study did not define ecological validity nor explain how performance connected to everyday functioning. Panerai et al. [58] compared VR-based training with traditional therapy but similarly offered little rationale for how VR task structure aligned with daily activity demands or with transfer measures. Across this body of work, stakeholder involvement is minimal, and ecological validity is often invoked but seldom defined, leaving design rationales opaque—limiting understanding of how VR can meaningfully support everyday functioning.

A notable exception is Foloppe et al. [21], who conducted a six-month case study in which a PwD practiced cooking tasks in a non-immersive VR kitchen and later replicated them at home. Although they did not explicitly invoke ecological validity, their design embodied verisimilitude by mirroring the participant's own appliances and cues and directly assessing real-world transfer. This work illustrates how meaningfully aligned VR tasks can support daily living. However, such examples are rare, and most IADL-based VR studies remain disconnected from the lived contexts and priorities of PwD and caregivers. These limitations highlight the need for VR systems that are informed by stakeholder perspectives and grounded in everyday routines, capturing the situated cognitive and functional demands of real-world activities over decontextualized cognitive measures.

2.3 Design Gaps in HCI/CHI: From Rhetoric to Reality

While rehabilitation-focused VR often struggles to connect with the values and priorities of PwD, HCI VR for dementia takes a different orientation: supporting personhood through recreation, reminiscence, and emotional engagement. Recreational systems promote creativity and social connection [11, 20, 29, 64], while reminiscence VR seeks to evoke autobiographical memories and reinforce identity [71]. Exergames encourage physical and cognitive stimulation through playful interaction [18, 35]. These approaches often draw meaningfully on lived context, making participatory and co-design practices especially important [1, 29, 42, 64, 75].

Yet, despite this emphasis on participation, PwD are frequently only peripherally involved in VR design. They may be observed during early empathizing sessions or included for late usability testing, while caregivers or clinicians influence key conceptual decisions [18, 35, 45, 71]. This mismatch between participatory rhetoric and practice limits insight into what PwD themselves find relevant or meaningful, and proxy-led decisions often prioritize safety at the expense of independence [43]. Hodge et al. [29] describe this dynamic as a *protectionist paradigm*, urging more nuanced forms of risk-taking in dementia-related design. Although rich in affective and relational value, these systems seldom extend to everyday functional activities, limiting guidance on designing VR that meaningfully supports daily living for PwD.

2.4 Emerging Intersections and Opportunities for HCI/CHI

While still limited in scope, recent work in HCI has begun to tentatively explore everyday functional activities more directly, often drawing on occupational therapy perspectives [15, 16, 75]—person-centered approach that emphasizes supporting IADLs within PwD's environmental, social, and personal contexts [37, 44]. Kim et al. [36] illustrate this emerging direction through an IADL-inspired food-ordering task designed for early detection of mild cognitive impairment. However, their system reflects challenges common in neuropsychology-driven VR, including limited stakeholder input, a developing design rationale, and initial references to ecological validity that remain to be fully articulated. Although such projects gesture toward interdisciplinary integration, they reveal the need for clearer conceptual grounding to guide future work.

Dole and Ju [17] provide the most explicit discussion in CHI of how VR experiences should correspond to real-world performance, though their analysis is situated in a search-and-rescue context rather than rehabilitation. They argue that veridicality—predictive accuracy—is often infeasible or unethical to evaluate in early-stage VR systems and instead propose **face validity** as a practical design alternative: whether the VR experience appears believable and appropriate for its intended purpose. They further link face validity to **realism**, understood as the coherence between virtual environments and their real-world counterparts in terms of appearance or function. Applied to IADL-focused VR for dementia, this perspective highlights how perceptions of believability and functional plausibility shape whether PwD and caregivers view tasks as realistic or useful.

Together, these limitations help explain why VR explicitly designed to support PwD's everyday activities remains rare. Existing systems emphasize assessment [3, 33, 65], rely on minimal stakeholder input [3, 21, 33, 51, 58, 65], or repurpose tools developed for other populations [51]. Ecological validity is frequently invoked but inconsistently defined, often conflated with immersion or visual fidelity rather than grounded in functional task demands. Recent conceptual work provides promising foundations. Suchy et al. [74] highlight verisimilitude—how closely VR tasks reproduce the functional and cognitive demands of real-world activities—as especially salient for PwD, while Dole and Ju [17] offer face validity as a design-actionable complement. Building on these insights, our study clarifies how ecological validity should be understood for IADL-focused VR and identifies how systems can reflect the lived routines and caregiving contexts of PwD to support meaningful everyday engagement.

3 Methods

Our goal in this study is to investigate how VR can be designed to support PwD in their daily living tasks. Specifically, we sought to understand: (1) how PwD, caregivers, and OTs define realistic and useful VR-based support for everyday activities and (2) how VR environments can achieve ecological validity by aligning with lived experiences and daily priorities. To address these questions, we adopted a two-stage interview design to separate exploratory sense-making from evaluative and generative activities. Each stage consisted of semi-structured interviews with PwD, caregivers, and occupational therapists (OTs), and we used visual probes in the second stage to ground the discussion of potential VR-based IADL practice. This study was reviewed and approved by the Institutional Review Board of Indiana University (#20919).

3.1 Participants

Recruitment took place between January 2024 and April 2025. We recruited PwD and their caregivers through outreach at adult day centers, personal connections, fliers distributed through community organizations, and events such as memory cafés [4]. We also employed remote-focused strategies, including posting on national dementia research registries and contacting advocacy groups, which helped expand participation to rural areas and lower socioeconomic groups facing barriers to in-person engagement. We recruited OTs through professional connections and referrals. We recruited additional participants following snowball sampling [59].

Our inclusion criteria required OTs to be licensed and experienced in dementia care; caregivers to be actively involved in essential daily care; and PwD to have a formal diagnosis of dementia or a related condition, live at home (to focus on an understudied population compared to community-living recruitment), and be able to communicate with researchers and provide informed consent. All participants provided informed consent prior to participation, and consent was reaffirmed at the beginning of each session. Participants had the option to discontinue at any time.

We interviewed a total of 11 PwD, nine caregivers, and eight OTs. PwD participants (six females, five males) had an average age of 69.3 years (SD 10.2; range: 53–88) and had been diagnosed on average 6.6 years prior (SD 2.7; range: 3–12). We present detailed PwD demographics in Table 1. Caregivers (eight females, one male) had an average age of 58.5 years (SD 15.9; range: 31–75) and were most commonly spouses (n=4) or adult children (n=4), with one non-family caregiver. Table 2 provides additional caregiver demographic details and their relation to PwD.

Some PwD and caregivers participated as dyads, meaning both members of a caregiving relationship were enrolled in the study. These pairs are indicated across Tables 1 and 2, where caregiver IDs are linked to the corresponding PwD IDs (e.g., P4–C2). Not all participants were enrolled as dyads: some PwD took part without their caregiver, and some caregivers participated independently of the PwD they support. OTs (six females, two males) reported an average of 15.3 years of professional experience (SD 12.8; range: 1–37). Table 3 summarizes OT demographics, education, and prior VR experience.

3.2 Procedure

The interview process was iterative. In the first stage, we conducted one-hour exploratory interviews with PwD and caregivers, focusing on: (1) lived experiences with dementia and its impact on daily activities, (2) meaningful tasks and routines in everyday life, and (3) strategies and adaptations used to maintain independence. We also interviewed OTs to gain a deeper understanding of their experiences and perspectives on: (4) OT practices and tailoring interventions for PwD, (5) considerations for VR task design, such as structure, adaptive difficulty, and skill transfer, and (6) strategies to support usability and ecological validity.

Insights from these interviews, together with existing literature, informed the development of initial sketches and a storyboard depicting a kitchen and grocery store. These visual probes were used to ground the discussion by providing participants with concrete environments to react to and modify, allowing them to identify which objects, layouts, cues, and activities felt realistic, what they would add or remove, and how practicing these tasks in VR could support their everyday routines.

These materials (Fig. 1, Fig. 2) were then used in subsequent one-hour interviews as visual probes for brainstorming [7, 77]. The storyboard depicts a smartphone-operated headset as the VR option, selected to prompt discussion of a potentially more accessible and affordable at-home setup. This choice was informed by prior literature highlighting cost and everyday applicability as critical factors for adoption [25, 27]. In these second-stage interviews, we invited participants to comment on task relevance, interaction sequences, complexity, and the potential usefulness of VR-based support. To acknowledge their time and contributions, participants who completed the first round of interviews received a $25 Amazon gift card, and those who participated in the second round received a $50 Amazon gift card.

We adopted this two-stage structure to first elicit participants' everyday experiences and then build on these insights using visual probes. This approach allowed participants to ground their reflections in concrete environments and articulate design-relevant details that would have been difficult to express abstractly.

Table 1: Person with dementia (PwD) demographics. M: Male, F: Female. NH: Non-Hispanic. "–" denotes information that participants chose not to disclose.

ID	Age	Gender	Race/Ethnicity	Marital Status	Primary Caregiver	Current Diagnosis	Years since Diagnosis
P1	53	F	Black, NH	Married	Spouse & Child	MCI & Moderate Memory Loss	8
P2	58	F	White, NH	Married	Spouse	PCA	4
P3	63	M	White, NH	Married	Spouse & 2 Children	Early Onset Alzheimer's	6
P4	–	M	White, NH	Married	Spouse (C2)	General Dementia	5
P5	71	M	White, NH	Married	Spouse	General Dementia	9
P6	80	F	White, NH	Widow	Child (C5)	General Dementia	5
P7	67	F	White, NH	Married	Spouse	Early Onset Alzheimer's, FTD	12
P8	70	M	White, NH	Married	Spouse (C6)	FTD	5
P9	75	F	White, NH	Divorced	Child	Evolving Dx (suspected FTD)	3
P10	68	M	White, NH	Married	Spouse (C9)	Early Onset Alzheimer's	9
P11	88	F	White, NH	Divorced	Child	MCI	7

Table 2: Caregiver demographics and relation to persons with dementia (PwD). M: Male, F: Female. NH: Non-Hispanic. "–" denotes information that participants chose not to disclose.

ID	Age	Gender	Race/Ethnicity	Relation to PwD
C1	31	F	Black, NH	Child
C2	–	F	White, NH	Spouse (P4)
C3	69	F	White, NH	Non-family
C4	75	F	White, NH	Child
C5	50	F	White, NH	Child (P6)
C6	61	F	White, NH	Spouse (P8)
C7	–	M	White, NH	Child
C8	–	F	White, NH	Spouse
C9	65	F	White, NH	Spouse (P10)

3.3 Data Collection and Analysis

All interviews were conducted either in person or via Zoom, depending on participants' location. Local participants were interviewed in person, while those living out of state joined via Zoom. Each session lasted approximately one hour and was audio-recorded with participant consent. Recordings were subsequently transcribed verbatim. Following automated transcription, each transcript was reviewed, and errors were manually corrected to ensure accuracy.

We conducted an inductive thematic analysis [8], allowing codes and themes to be generated directly from the data. All authors individually reviewed the interview transcripts to ground themselves in the dataset and contribute to early open coding. As the first author had conducted the majority of interviews and was most familiar with the data, they synthesized these preliminary codes into a draft codebook. The second, third, and last authors then reviewed this codebook, contributing additional codes and suggesting refinements based on their own readings of the data. Through iterative meetings and follow-up email exchanges, the team collaboratively reviewed and refined this codebook, discussing disagreements, clarifying code boundaries, and iteratively reaching consensus on a shared coding scheme. Due to the exploratory and inductive nature of our analysis, we did not calculate inter-coder reliability; instead, we relied on team discussions to support analytic rigor [9, 12, 56, 57]. Building on this foundation, all authors carried out axial coding, through which broader subthemes and themes were iteratively developed, reviewed, and refined. In the following section, we present these themes in detail, illustrated with representative quotes from participants.

4 Findings

Across the four themes, we show how participants defined what makes VR realistic and useful for supporting IADLs. Each theme highlights critical elements for reflecting the lived contexts and priorities of PwD and caregivers: functional fidelity, meaningfulness, relational caregiving dynamics, and adaptability. Together, these themes demonstrate how ecological validity is grounded in participants' everyday experiences rather than in the immersive or visual fidelity of VR.

Table 3: Occupational therapist demographics and VR experience. M: Male, F: Female. NH: Non-Hispanic. MS: Master of Science in Occupational Therapy (or related field, e.g., Gerontology); OTD: Occupational Therapy Doctorate. MCI: mild cognitive impairment. *Past VR Experience* categories refer to prior exposure in either clinical or personal contexts.

ID	Age	Gender	Race/ Ethnicity	Education	Years of Experience	Area of Care	Primarily Working With	Past VR Experience
OT1	54	F	White, NH	MS	25	Outpatient Rehabilitation & Community Dementia Program	MCI–Moderate Dementia	None; familiar with concept
OT2	60	F	White, NH	OTD	37	Rehabilitation & University OTD Program Director	MCI–Moderate Dementia	Used clinically and for teaching
OT3	42	F	White, NH	MS	18	Driving Rehabilitation	MCI–Moderate Dementia	Limited; only personal exposure
OT4	28	M	Asian, NH	MS	2	Acute Rehabilitation & Skilled Nursing (Locked Dementia Unit)	MCI–Severe Dementia	Limited; only personal exposure
OT5	38	F	White, NH	MS	12	Acute Rehabilitation	MCI–Severe Dementia	Used recreationally, not clinically
OT6	43	F	White, NH	MS	23	Acute Rehabilitation & Home Health (Program Lead)	Moderate–Severe Dementia	Used clinically but infrequently
OT7	35	M	White, Hispanic	MS	4	Outpatient Rehabilitation	Moderate Dementia	Used clinically for stroke rehab only
OT8	28	F	Asian, NH	MS	1	Home Health	Moderate–Severe Dementia	Limited; only personal exposure

4.1 Beyond Replication: Functional Fidelity Over Visual Fidelity

Participants, especially PwD, challenged a widespread assumption in prior IADL-based VR work: that VR's immersion and visual fidelity provide higher verisimilitude than paper-based clinical instruments, and are therefore ecologically valid. This assumption rests on the idea that faithfully replicating a person's real-world environment will naturally reflect their everyday functioning. Yet, our findings suggest that even if VR reproduced a PwD's living space with high visual fidelity, it would not necessarily support IADL practice. Many participants emphasized that this assertion is not synonymous with functionality, safety, or clarity, and in some cases, replicating actual environments risks reinforcing dysfunction, confusion, or frustration. These accounts illustrate that for PwD, verisimilitude hinges on the cognitive and procedural demands of IADLs—not their surface appearance.

While the organization and familiarity of a PwD's environment can influence IADL performance, participants emphasized that executive function[1] remains the primary challenge, posing significant obstacles. For instance, P5 described how a familiar cooking task in his real kitchen quickly broke down due to a brief disruption in sequencing:

> *"I had all these ingredients out, lined up left to right... Then the phone rang. When I came back, I did them in reverse. I ended up putting in twice as much of one thing and not enough of another... My executive functioning is where I'm having the most struggle... It's hard to get things in order and to understand that you have to stir this before you add that."*

Others described how home adaptations, such as organizing spices or keeping countertops clear, can help reduce cognitive load during cooking.

Although a supportive environment can aid task completion, and modifications may be beneficial, participants cautioned that replicating a space that has not been optimized for dementia could recreate the very conditions that hinder participation in IADLs, rather than promote strategies or changes that make tasks more manageable. As P7 emphasized, environmental clutter is an active barrier to IADLs, noting that *"The more clutter, the more confusion... Why encourage them to maintain the clutter when you know it's going to increase their confusion?"*

Many participants also felt that effective IADL training hinges on functional fidelity rather than perfect visual replication. P10, for example, downplayed the importance of aesthetic detail, explaining

[1]Executive function refers to cognitive processes—including planning, working memory, inhibitory control, and cognitive flexibility—that enable goal-directed behavior and the regulation of thoughts and actions.



Figure 1: Kitchen and grocery store environment sketches used as visual probes during interviews.

that skills would transfer to real life as long as core functional elements were present:

> *"I don't think it [visual replication] would [matter]. I would carry [what I learn] over from the most simple kitchen... the idea of having a sink, a stove, a microwave, some cabinets, and a refrigerator—it doesn't matter where they are, doesn't matter if it's a 1950s rounded old refrigerator or what, they both serve the same purpose."*

At the same time, participants distinguished between superficial appearance and functional properties that shape safety and how tasks are carried out. P10 highlighted appliance type as a critical example:

> *"Electric versus gas... there's a big difference in cooking on the two and what can happen if you leave gas with a pilot [light] that's not lit."*

OTs echoed this emphasis on functional elements and task sequencing as the basis for carryover. As OT8 explained, training did not need to take place in a participant's own kitchen so long as key elements and action flows were preserved in VR:

> *"It doesn't have to be in their kitchen, but as long as the fridge is [there], or certain ingredients are right there [in VR]... That still transfers to when they're at home [in real-life]—they'll think, 'What did I do first? I looked for a cutting board,' and then they'll do the same."*

Taken together, these examples highlight that verisimilitude in VR arises not from reproducing a space visually, but from mirroring the sequencing, decision-making, and other executive function demands of real-world activity. Participants further emphasized that environments do not need to be visual replicas of any specific real-world settings; instead, they simply need to be cohesive and appropriate for the task (e.g., a kitchen with a countertop and the essential appliances relevant to cooking). Functional fidelity was also described as critical not only to the realism of practicing IADLs in VR but also to its usefulness: tools and appliances should operate as they do in a person's real life.

Thus, executive function emerged as the foundation of verisimilitude for participants, while context and functional fidelity underpinned perceptions of realism; together, these elements shaped whether VR is considered realistic and useful for IADL practice.

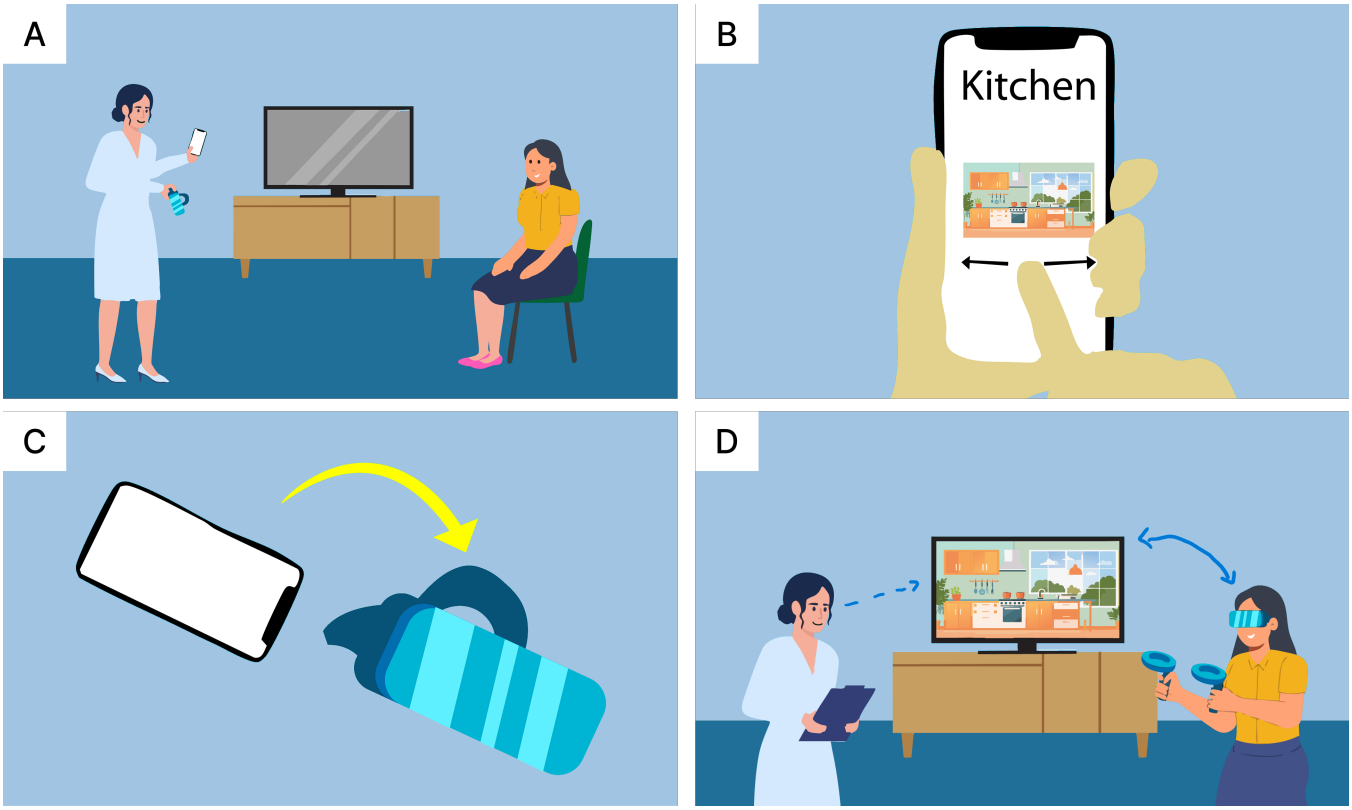


Figure 2: Storyboard illustrating the setup and use of a potential IADL-based VR at-home program. (A) An OT (or caregiver) prepares the system while the participant is seated. (B) The OT selects a relevant daily living environment (e.g., kitchen) from a smartphone interface. (C) The smartphone is placed into a lightweight headset to enable immersive interaction. (D) The participant engages in the VR activity using the headset and controllers, while the OT observes and provides support as needed.

4.2 Personal Meaning as a Driver of Engagement

Meaningful engagement emerged as a powerful driver of participation in IADLs. For many PwD, the value of a task lies not in its novelty or visual detail, but in its personal relevance: activities that aligned with their preferences, reinforced a familiar familial or household role, or allowed them to contribute in ways that felt useful. These activities were often intertwined with cultural background, gender norms, and a sense of purpose derived from helping others.

For many participants, the activities they valued most were shaped by a combination of cultural traditions and gender roles. Based on their experiences, OT7 described how, in the Hispanic community, cooking often held special significance, tied to family heritage and expectations:

> *"In the Hispanic community, a lot of patients and caregivers say cooking tends to be the main IADL that they enjoy doing... Some will do laundry, fold towels, and make the bed. Those household chores are usually—I guess for lack of a better term—matriarchal. So they did that with their mothers, and that's what they taught their daughters. And so that was their norm: to be housewives."*

These patterns were echoed more broadly, with gendered divisions of labor influencing the types of activities participants wished to maintain. For example, P8 expressed no interest in cooking, a task his wife had always managed, but described wanting to maintain engagement in the IADLs that reflected his traditional household role, such as minor car maintenance, changing lightbulbs, or assembling new furniture.

For some participants, the activity itself was both a genuine source of enjoyment and a link to family traditions. P1, for example, loved cooking and took pride in preparing large meals for holiday gatherings. She adapted her process to accommodate cognitive changes, finding strategies that allowed her to continue doing what she valued:

> *"Cooking is one of the things I love to do, and I still do it now. Some recipes are too complex, too many steps... I do what I can do... [I need] every step [of a recipe, step-by-step]... Sometimes when I make it, I'll watch a YouTube video, and then I'll pause, do that step, pause, do the next step, pause. And then, if I'm writing it out, I'll write it out step by step, my own way, separating*

> *the steps. Not the paragraph format. That's confusing for me."*

Similarly, P7—though she admitted she does not enjoy cooking—made significant efforts to restructure a cherished family recipe, her mother's macaroni and cheese, after she began struggling to remember the sequence. Rather than give it up, she created a laminated, step-by-step checklist and uses a dry-erase marker to track her progress. Reflecting on this process, she described how VR could support repeated practice and reduce her reliance on external aids:

> *"I think that's where VR can come in handy is to get me to practice it... Maybe I wouldn't need this sheet so much anymore."*

For both P1 and P7, the goal was not to replicate former routines exactly but to adapt them in ways that preserved the meaning of the activity while making it achievable. P7 framed VR as supporting a process of re-learning and re-anchoring actions through visual, hands-on experience, explaining that VR offers *"the joy of seeing something, not just hearing it or reading it."* She contrasted this with written instructions, noting that *"when I'm reading... I now have to take what I read and put it into visuals... If it were in virtual reality, [I]'d be able to visualize it a lot better."*

OTs stressed that this personal meaning was not only emotionally valuable but directly tied to adherence. As OT2 explained, *"I think if the program isn't meaningful to them, they're not going to do it."* Tailoring activities to individual interests was a recurring strategy for sustaining engagement. OT6 described reframing therapy for a retired nurse with dementia, who disliked conventional exercises and puzzles, by integrating clinical problem-solving scenarios drawn from the patient's nursing background, which kept her patient *"engaged in something she actually enjoys."* In another case, a patient resisted traditional exercises but enjoyed coloring. OT6 used that interest as an activity for fine motor and wrist movement:

> *"[One of my patients] battled her caregiver on exercises, but she liked coloring. So [I had her] color each day with colored pencils to work fine motor, wrist, and other movements. [Then,] I saw [improvement]—'Okay, you've got most of your range back... [Now, we can] start doing dishes and [other activities].' We've done your exercises without you doing [traditional] exercises because you hate [them]."*

Collectively, these accounts suggest that IADL participation may not be driven solely by abstract clinical goals, but by activities that feel meaningful. When the VR task and environmental context reflect PwD's cultural identity, personal history, or valued household roles, they are perceived as relevant and believable, thereby enhancing realism. As a result, practice becomes something they want to return to, increasing the likelihood that skills developed in VR will transfer into real-world contexts—central to both the usefulness and veridicality of IADL practice in VR.

4.3 Leveraging VR for Collaborative Caregiving

Across interviews, caregivers emerged as a critical component of PwDs' lives, shaping what IADLs are performed, where they take place, and how they unfold in practice—defining the realistic context and functional fidelity of everyday activity. They are active collaborators, co-performing tasks, interpreting signs of struggle, and adapting strategies to sustain participation. OTs highlighted their role in interventions, describing them as key contributors in personalizing therapy and reinforcing occupational strategies to encourage skill improvement and carryover at home.

Caregivers also played a crucial role in PwD's motivation. For instance, OT5 emphasized the value of a caregiver's presence in interventions, explaining how *"it's so much better when you have a family member or a caregiver there because they know... and trust that person."* Caregivers frequently adapt home environments and daily routines in ways that parallel, and sometimes exceed, what VR could potentially offer. These adaptations range from logistical planning to environmental modification. For example, C9 uses a meticulously organized grocery list to make shopping more efficient, describing *"Excel spreadsheets with our grocery list in the store... [ordered] by the aisles."* Many described cooperative approaches to IADLs that distribute responsibilities according to abilities. In cooking, C9 and P10 have developed a shared workflow:

> C9: *"I do the actual chopping and mixing and prep and all that, but he... will get stuff for me out of the fridge... He does that part."* P10: *"I try to make her job a lot simpler, and I'm an active part of it."*

This task division offloads higher-order cognitive demands to the caregiver while preserving meaningful participation for the PwD, shaping the verisimilitude VR must support.

In contrast, other caregivers err too much on the side of caution, limiting PwD's autonomy. Some OTs cautioned that overprotection, while well-intentioned, may inadvertently undermine therapeutic goals. OT1 asserted, *"there has to be some risk in all of our lives,"* likening participation in IADLs to exercise as *"the best treatment for blood pressure."* OT7 described similar feelings and considered how caregiver burden could even be reduced:

> *"They want [the PwD] to be as far away from the kitchen as possible... I try to help [the caregiver] understand... the [caregiver burden] can be managed a little more if you allow a little more autonomy where it can be allowed."*

In other cases, OTs reported working with caregivers who needed prompting to implement environmental changes before accidents occurred. OT4 suggested that using VR could act as a *"wake-up call,"* while OT5 saw potential for resolving disagreements:

> *"[We can] simulate this, this is what's going to happen, take a look... then it can be very obvious in certain cases that safety is a concern."*

These examples demonstrate how caregivers are already creating adaptive, flexible support at home; however, OTs warn that caregivers should find an appropriate balance between protecting PwD from harm and allowing them the independence to engage in meaningful activities. This balance was a recurring theme across interviews, with OTs noting that caregiver decisions often shaped not only the safety of IADL participation but whether those activities occurred at all.

These tensions were further shaped by cultural expectations around caregiving. Some OTs observed that in some cultures, providing care often takes precedence over fostering independence. For instance, OT5 explained how there's an *"expectation that the patient will be taken care of. There's not always an emphasis on [them] being as independent as possible."* OT7 specifically noted this with his Hispanic patients:

> *"The Hispanic community—I will say, just the culture itself—it's very much like, 'We take care of our parents because they took care of us.' It's a kind of 'give back' culture."*

These relational dynamics indicate that IADL engagement is not solely an individual phenomenon; instead, it emerges through the caregiving ecosystem that structures who perform which IADLs, how responsibilities are negotiated, and what kinds of participation are considered appropriate or safe. Consequently, VR realism must reflect this context and functional organization, while verisimilitude must capture the executive demands as they are distributed across caregiver–PwD collaboration.

Several OTs noted that VR could help caregivers better understand a PwD's true capabilities. OT7 envisioned VR as a tool to *"help bring out that light,"* and described how *"advocating for their autonomy can be facilitated through the virtual experience."* These views suggest that VR has the potential to act as a neutral ground where both PwD and caregivers can explore safe participation, practice risk management strategies, and build mutual understanding of what is both possible and safe. It also offers a useful alternative to existing safety-awareness tools, which many OTs critiqued—for example, OT6 referred to hazard cards *"from the 70s"* that bear little relevance to contemporary risks, or in-clinic simulations where hazards are obvious because PwD have seen therapists set them up in advance. When hazards are represented realistically, VR can make risks feel realistic, support constructive conversations between PwD and caregivers, and encourage proactive environmental changes without immediately stripping away autonomy.

In addition to negotiating safety and autonomy, participants emphasized that VR could support collaboration by accommodating multiple caregiving roles, such as serving as in-session collaborators, post-session coaches, interpreters of progress, or observers who use insights to adapt the home environment and reinforce practice. Embedding this flexibility into VR design was seen as central to its usefulness, as it allows systems to mirror the relational nature of everyday IADLs and support more responsive caregiving partnerships.

Together, these perspectives show that designing VR for collaborative use supports realism, preserves verisimilitude, and enhances practical usefulness.

4.4 Toward Dynamically Adaptive VR

Participants envisioned VR not as a static simulator of real-world environments, but as a dynamic platform for scaffolding participation in daily activities. They emphasized the need for both horizontal adaptability (across users) and vertical adaptability (for the same user) over time. This adaptability was positioned as foundational to what makes VR feel realistic and useful, and to preserving verisimilitude, because dementia symptoms can fluctuate from day to day, requiring the system to respond in real time to changes in cognitive and functional capacity.

A recurring theme was the value of presenting IADLs as clear, sequential actions, with adjustable complexity to match current capabilities. Several participants described adapting daily routines this way. For example, P3 explained that receiving instructions *"step-by-step"* made tasks more manageable, noting that when his family breaks meals down incrementally, *"that makes it easier on me because I can remember to add this or add that or take it away or something. I can do that."* P10 applied a similar approach to cooking, finding that preparing and measuring ingredients in advance made recipes easier to follow:

> *"When you're preparing a meal... get out as many ingredients as you can and measure them out in advance... When it comes to the actual recipe, I can follow it better... Whereas most recipes... that's not how somebody with dementia looks at things or that's helpful."*

OTs also stressed the importance of dynamically adjusting task difficulty to maintain the *"just-right"* challenge—scaffolding upward as skills improve and simplifying when needed. Participants noted that abilities can shift dramatically from one day to the next, making flexibility essential. To address this, OT7 recommended real-time adjustments supported by concrete cues:

> *"[Features I'd want control over?] The choice of being able to downgrade and upgrade in vivo... having [clear] instructions... 'this pan' or 'the red bowl.' Maybe having arrows pointing to things, just giving them those cueing labels. I think labels would be good, because that's something I usually recommend with patients."*

OT8 suggested a staged approach, starting with a distraction-free kitchen and gradually adding background noise or extra items as skills improved. OT7 connected this idea to clinical practice, where tasks are progressively expanded from one or two steps to multi-step sequences.

Participants emphasized that long-term engagement in IADL-based VR relies on both variety and encouragement. OT7 noted that engagement can plateau if tasks are not varied or progressively challenging, explaining that *"if they get good at something, I have to change it up for them... even if they don't complete something, maybe they get bored while they're working toward a goal."* To maintain momentum, OT7 suggested features such as a user-initiated *"next"* or *"skip"* button to preserve autonomy when a task becomes too complex. He also emphasized the role of encouraging, context-sensitive feedback, noting that it should be *"just to a point where it's not condescending, but motivating."*

This encouragement extended to how mistakes were handled. Rather than punitive responses, participants preferred feedback that supported learning and confidence. P7 valued timely cues that prevented errors before they happened, while P9 described gentle prompts that encouraged self-correction. Others, like P10, felt it was better to make the mistake in VR and then receive feedback that turned it into a teachable moment:

> *"Let it actually happen... then [alarm sound], 'No, you made a mistake.' Ask what mistake you think you made, and if you can't figure it out, explain and have them*

try again... Make it a learning experience, not just an awareness that you goofed."

P1 added that timing is as important as tone: Feedback should allow space for processing and avoid escalating frustration. These insights illustrate the need for feedback that is accurate, actionable, and tailored to the individual's preferred learning style—delivered with the right balance of immediacy, tone, and respect for cognitive processing needs.

These patterns reveal that ecological validity is dynamic: because PwD's cognitive and functional abilities fluctuate from day to day, VR must adapt in real time to maintain real-world executive function demands. Participants also viewed IADL-based VR as a bridge between present abilities and the activities they hoped to maintain or regain. Adaptability, variety, and affirming feedback were seen not as optional features but as essential for aligning the virtual experience with fluctuating capacity, supporting realism through functional fidelity, and sustaining usefulness through meaningful participation. By incorporating relevant IADLs, adjusting to changing needs, and reinforcing successes, VR can both respect current challenges and support future goals. Adaptable sequencing, graded difficulty, and supportive feedback allow VR tasks to mirror real-world performance and sustain verisimilitude as abilities evolve.

5 Discussion

Our findings show that perceptions of *realistic* and *useful* VR for everyday activities hinge on how well systems align with multiple, intersecting aspects of PwD's everyday lives. Participants treated VR as realistic when it reflected the functional and cognitive demands of daily tasks, resonated with personal and cultural identities, aligned with caregiving routines, and adapted to changing abilities. These insights challenge design approaches that prioritize visual replication over the lived contexts that shape PwD's participation. In the sections that follow, we show that while functional fidelity and verisimilitude are foundational to IADL-based VR, they must be integrated with caregiving dynamics, personal meaning, and aspirational goals to effectively support IADL practice at home.

5.1 Functional Fidelity and Verisimilitude as the Foundations of IADL-Based VR

Our findings showed that participants consistently emphasized the cognitive and functional demands of real-world tasks as the most important aspect of IADL-based VR. Specifically, executive function—planning, sequencing, multitasking, and error recovery—emerged as the primary anchor for verisimilitude. Participants described VR as realistic and believable when the virtual environment supported verisimilitude through **functional fidelity**, by providing the appropriate tools for the task and ensuring those tools behaved as their real-world counterparts would. When VR upholds these foundational elements, the structure and decision points of familiar IADLs are preserved, making PwD more likely to act as they would in daily life—an indicator of **presence**, or the extent to which users treat virtual environments as though they were real. When presence is achieved, it not only offers important insights for OTs and caregivers, but also increases the likelihood that IADL skill gains will transfer to real-life performance [2, 10, 14, 26, 72], which ultimately shapes what makes VR useful.

In contrast, prior IADL-focused VR systems have frequently treated ecological validity as a matter of visual replication or hardware-driven immersion [3, 33, 60, 65, 74]. These approaches assume that detailed textures, realistic lighting, or 360° environments will naturally translate into greater engagement or realistic behavior [33]. Our findings push back against this assumption, asserting that executive function underpins verisimilitude. Participants further emphasized that realism and clinical usefulness depend less on visual detail than on whether a system captures the functional distinctions and task logic that matter for everyday living. Designing a *cooking task*, for example, requires attending to what PwD typically prepare in their daily lives, which appliances they use, how task steps should be sequenced to surface executive-function breakdowns, and how cues and tools should respond to user actions. Treating ecological validity as a visual property obscures these practical considerations and risks creating VR tasks that look real but do not reflect real-life functioning.

The need to support functional fidelity becomes even more apparent when considering the wide variability of household tools and appliances. For example, IADLs as seemingly straightforward as doing laundry can vary dramatically depending on the washer model: some machines rely on a single dial and a few buttons, while others use touchscreen menus with dozens of cycle options. Even the sequence of steps—selecting a cycle, adding detergent, and initiating the machine—differs across models. This variability raises a design question central to ecological validity: which aspects of interaction must remain faithful to the user's real-world context, and which can be generalized without compromising verisimilitude? Our findings suggest prioritizing fidelity for elements that meaningfully affect task performance—such as safety-relevant behaviors, sequencing dependencies, and opportunities for executive-function breakdown—rather than around literal replications of entire environments.

To address this challenge, participants and therapists pointed to the value of personalization mechanisms that align VR tasks with the user's real-world appliances, routines, and skill levels. Advances in AI offer one promising pathway for achieving such alignment at scale. Prior work [81] demonstrates how AI can generate or adapt virtual environments from text or image inputs, reducing the burden of manual modeling. This approach could be applied in VR to integrate individualized tools and layouts that reflect users' homes. For PwD, this alignment may reduce cognitive load associated with unfamiliar objects and support transfer to real-world performance. For therapists and caregivers, AI-assisted customization offers a practical way to tailor training without extensive technical effort.

Design Implication. VR systems should therefore (1) foreground executive-function demands by structuring tasks around realistic sequencing, decision points, and error opportunities; (2) prioritize fidelity to the functional behaviors of tools and appliances over literal environmental replication; and (3) support scalable personalization—potentially through AI-assisted customization—to reflect the heterogeneity of users' home contexts.

5.2 Ecological Validity as Co-Constructed Through Caregiving

Our findings showed that everyday IADLs for home-dwelling PwD are not individual tasks but shared activities structured through longstanding caregiving relationships. PwD and caregivers often enact IADLs together, coordinating responsibilities in ways that maintain participation while managing cognitive and physical challenges. This collaborative dynamic reframes ecological validity as something co-constructed, since executive function demands and task structure are shaped by how responsibilities are shared between PwD and caregivers. These insights extend prior work on the relational nature of dementia care [68, 75], showing that a VR task cannot feel realistic or useful unless it reflects the caregiving patterns that organize how the activity unfolds in daily life.

A recurring expression of this collaboration was a negotiated division of labor shaped by safety, efficiency, and personal preference. PwD typically performed steps aligned with familiarity or lower-risk actions, while caregivers assumed responsibility for tasks involving hazards or greater cognitive load. These patterns were stable, tailored, and deeply embedded in lived routines.This relational structuring of IADLs stands in contrast to how IADLs are commonly framed in existing VR systems for dementia, which are largely developed in institutional or community-based care settings where activities are staff-directed and caregiver–PwD relationships are formal rather than familial [78]. Our findings show that IADLs in the home cannot be abstracted from these intimate caregiving dynamics: what activities PwD engage in, how they participate, and what those activities mean are co-shaped by their caring ecosystem.

Supporting this collaborative structure in VR raises key design considerations. Personalization should emphasize the actions PwD habitually perform while also acknowledging caregiver-led steps that shape the overall task flow. One promising direction is to support shared or co-operative modes in which PwD and caregivers can jointly inhabit a virtual task environment. Prior work shows that caregiver presence in VR can bolster motivation, reassurance, and persistence [19, 35]. However, our findings also indicate that such direct involvement is not always feasible. Caregivers often supervise from the periphery or engage intermittently due to competing demands, suggesting the need for multiple modes of caregiver involvement—from direct task participation to lightweight observation or timely assistance. This emphasis on flexible participation aligns with prior calls for systems that balance caregiving workload with opportunities for shared engagement within care dyads [16, 31, 43].

Design Implication. VR systems should (1) reflect collaborative task structures by modeling typical PwD and caregiver actions; (2) support multiple caregiver-involvement modes, including co-performance, remote monitoring, and intermittent assistance; (3) foreground task elements PwD typically lead, while representing caregiver-led steps in ways that preserve task coherence; and (4) provide interfaces that allow caregivers to step in or out as needed.

5.3 Negotiating Safety and Risk: Representing Hazard Awareness in IADL-Based VR

Our findings also showed that caregiving relationships played a central role in how safety was interpreted, negotiated, and managed during IADLs. Caregivers frequently modulated PwD participation based on perceived risk, determining when to intervene, supervise, or allow independence. Yet OTs emphasized that perceived risk did not always align with actual risk: some caregivers restricted participation unnecessarily, while others underestimated hazards with meaningful implications for safety. This divergence mirrors longstanding observations that dementia care oscillates between overprotection and underestimation of risk [5]. These dynamics show that for IADL-based VR to be useful, it must facilitate safety negotiations among PwD, caregivers, and OTs.

Participants viewed VR as a promising medium for surfacing these safety dynamics in a realistic yet controlled manner. Rather than relying on decontextualized hazard assessments commonly used in practice, embedding hazards directly into task flows allows PwD to encounter risks without physical consequences. Building on prior work by Sun et al. [75], Siette et al. [69], and our participants' accounts, we posit that this approach can surface naturalistic safety responses through presence, provide OTs with rich behavioral evidence to support more robust intervention planning, and facilitate shared safety awareness between PwD and caregivers.

Safety-related performance in VR raises challenges that extend beyond design. Assessing whether virtual behavior predicts real-world safety—traditionally understood as veridicality—is ethically fraught for high-risk IADLs, where errors cannot be safely reproduced. Similar challenges have been noted in Dole and Ju's search-and-rescue VR work [17], where real-world validation is infeasible. In such cases, *face validity* offers a practical alternative: whether a virtual scenario appears appropriate and believable for its intended purpose when real-world predictive testing cannot be ethically or practically conducted. More broadly, face validity provides an essential grounding for early-stage VR design, offering a way to evaluate plausibility and usefulness before real-world transfer can be safely or meaningfully assessed.

Designing for face validity in safety-critical IADLs requires that hazards, task demands, and system responses feel functionally plausible and cognitively coherent. Participants emphasized that dangers must be visible and interpretable. Concrete cueing strategies—including arrows, color highlights, simplified instructions, and gentle reminders delivered at moments of confusion or risk—were viewed as especially helpful. At the same time, hazards should convey safety consequences in ways that build awareness without causing undue distress. When carefully designed, these representations can foster a sense of presence and serve as a practical safety negotiating tool.

Design Implication. VR should (1) embed hazards within functional task sequences rather than decontextualized safety assessments; (2) calibrate risk visibility and cueing to enhance awareness without overwhelming the user; (3) surface naturalistic safety behaviors through a sense of presence that therapists and caregivers can interpret together; and (4) offer mechanisms for shared review, enabling PwD and caregivers to align their understanding of risk.

5.4 Personal, Cultural, and Identity-Aligned Meaningfulness in IADL-Based VR

Our findings showed that realism alone could not sustain engagement in IADL-based VR; activities also needed to feel meaningful to

PwD. Participants emphasized that they were far more motivated to practice tasks that aligned with their everyday roles, preferences, and valued routines. This emphasis resonates with rehabilitation research demonstrating that meaningful activities promote greater functional improvement and adherence [13]. Therapists similarly noted that interventions must be grounded in personally meaningful activities to support adherence, engagement, and therapeutic impact. Meaningfulness thus plays a central role in shaping whether VR feels both realistic and useful: without alignment to identity and personal values, even strong functional fidelity and verisimilitude are unlikely to sustain engagement.

Our findings further reveal that meaningfulness is shaped by cultural patterns that influence which IADLs feel familiar, worthwhile, or connected to heritage. Several OTs also noted that they worked with families from a range of cultural backgrounds, which shaped how different IADLs carried meaning for PwD. They described how certain activities—especially cooking—held particular cultural significance for PwD in Hispanic families, reflecting traditions transmitted across generations and expectations around family roles. These findings echo Nan et al. [50], who demonstrated how culture shapes the meaning of reminiscence experiences and highlighted the difficulty of avoiding Western-centric assumptions. In IADL contexts, such assumptions can reduce recognition or emotional connection when virtual tasks do not reflect culturally relevant foods, tools, or practices. Supporting cultural variability in VR—whether through recipe options, utensil types, or context-specific cues—may therefore be crucial for ensuring that IADL-based VR resonates with diverse users.

Gender norms also influenced the meaningfulness of daily activities. Women in our study frequently described cooking as part of their long-standing household responsibilities, while men more commonly identified with tasks such as home repair or maintenance. This mirrors prior work in dementia-focused VR showing gendered preferences in reminiscence experiences [71]. Therapists noted that cultural and gender expectations often intersected, shaping which IADLs carried personal meaning and which felt incongruent. Designing IADL-based VR around neutral, generic tasks risks overlooking these patterns and diminishing the relevance of the experience to users' lived identities.

Beyond cultural and gendered influences, meaningfulness was also deeply tied to personal identity. PwD described specific recipes, holiday dishes, or family routines that they were proud of and motivated to continue practicing. These activities carried emotional significance and connected them to long-standing roles within their families. This finding aligns with prior dementia-technology research highlighting the importance of supporting personhood and continuity of self [5, 29, 61, 64, 75]. Participants expressed enthusiasm for VR scenarios that would allow them to relearn or adapt cherished tasks—such as a favorite recipe—suggesting that meaningfulness extends beyond engagement and into self-affirmation, confidence, and motivation to practice.

Together, these insights demonstrate that meaningfulness in IADL-based VR emerges from multiple layers of experience: cultural practices, gendered expectations, and personal identity. Thus, realistic and useful VR cannot rely solely on generic task libraries or templates. Instead, it must reflect the richly varied contexts that shape what PwD view as worthwhile, recognizable, and motivating.

Design Implication. VR systems should (1) provide individual customization so that activities can be tailored to users' cultural backgrounds, preferred tools, and familiar household practices; (2) accommodate habitual routines by allowing task sequences and interaction patterns to reflect how users ordinarily approach, organize, and carry out everyday activities; and (3) support identity-linked activities that preserve personal history and sense of self.

5.5 Supporting Aspirational Goals Through Adaptive and Personalized IADL-Based VR

Our findings showed that PwD's motivation to participate in IADL-based VR was shaped not only by tasks they currently perform, but also by activities they hope to continue doing or regain the ability to do. Many participants described a desire to maintain meaningful roles—such as preparing a beloved family recipe, hosting a holiday meal, or managing familiar household routines—that were becoming difficult but still held deep personal significance. This reflects longstanding psychological and rehabilitation models emphasizing that long-term, aspirational goals can be powerful sources of intrinsic motivation [67]. Similarly, prior work in HCI and health technologies has demonstrated that systems designed around users' aspirational selves—not merely around their present abilities—can promote sustained engagement and effort [48, 66]. Yet this aspirational dimension has received little attention in VR for dementia. Our findings extend this work by showing that IADL-based VR must align not only with PwD's lived realities, but also with who they strive to be and what they hope to continue doing, opening new design directions for more meaningful and motivating VR experiences.

To support these aspirations, OTs emphasized the importance of tools that enable VR tasks to reflect personal routines, functional baselines, and changing needs over time. Participants valued the idea of longitudinal profiles that could store preferred activities, adaptive strategies, and performance histories. Therapists envisioned environmental templates that would allow quick reconfiguration of VR spaces—such as switching between handwashing dishes and using a dishwasher, or between electric and gas stoves—mirroring real-world choices and reducing setup time during sessions. This perspective aligns with Hodge et al.'s notion of flexible templates [29] and responds to broader calls for systems that document and visualize change over time to support data-driven intervention planning [1, 75]. Profiles could also embed individualized adaptations, extending work by Matsangidou et al. [46] that tailored virtual environments to patient preferences and personal histories. Incorporating features such as high-contrast tools or simplified layouts allow VR to better match PwDs' capabilities and preferences rather than imposing a one-size-fits-all structure.

However, personalization cannot be static. Our findings revealed that PwDs' abilities fluctuate from day to day, requiring VR systems that can respond dynamically to moment-by-moment performance. Some PwD preferred immediate cues that prevented errors before they occurred, while others found it more helpful to make mistakes and receive reflective feedback afterward. OTs similarly emphasized the need to maintain a just-right challenge, adjusting sequencing, complexity, and cueing to foster confidence without oversimplifying the task. Ideally, therapists would remain in control of these

adjustments, drawing on professional expertise and caregiver input. Yet practical constraints—geographic disparities, limited therapy hours, and the need for frequent practice—make it unrealistic to rely solely on therapist-driven customization.

These challenges highlight the need for models that extend rather than replace therapist expertise. A human-in-the-loop approach offers one promising direction, enabling therapists to oversee and refine algorithmic suggestions while reducing the workload of constant manual adjustment. Prior work demonstrates the feasibility of such adaptive systems: individualized profiles have been used to tailor VR task parameters based on performance history and error patterns [32], while demographic, preference, and physiological data have been leveraged to inform dynamic environmental adaptations [46]. Our findings suggest that adapting these mechanisms to IADL practice could help fine-tune difficulty and feedback in real time while preserving therapist authority to review, modify, or override automated adjustments. Caregivers may also participate in this oversight role, but interfaces must remain intuitive and low burden to support feasible everyday use.

Design Implication. VR systems should (1) incorporate longitudinal profiles that preserve preferences, routines, and performance histories; (2) enable flexible reconfiguration of environments; (3) provide real-time adaptive mechanisms; and (4) support human-in-the-loop oversight that balances algorithmic assistance with therapist and caregiver expertise.

6 Limitations and Future Work

This study provides an in-depth examination of how IADL-based VR can support everyday participation for PwD within home environments, but several limitations shape the scope of our findings and point to directions for future work.

Our use of sketches and a storyboard—rather than a fully functional VR prototype—enabled participants to reflect on everyday activities without being constrained by device usability or sensory demands. However, this approach could not capture embodied interaction, ergonomics, or real-time feedback dynamics that may influence how PwD respond in immersive contexts. Future work should evaluate these design considerations using working HMD systems to understand how embodied experience shapes perceptions of usefulness, realism, and safety.

Our sample focused on home-dwelling PwD supported primarily by family caregivers. This context offered valuable insight into the relational nature of IADLs and the caregiving practices that shape daily activity, yet it may not generalize fully to facility-dwelling PwD who interact with staff rather than long-standing family partners. At the same time, many themes we identified—such as fluctuating abilities, the importance of task relevance, and the need for clear scaffolding—may extend to other neurodegenerative populations living at home (e.g., Parkinson's disease, Amyotrophic Lateral Sclerosis, multiple sclerosis), who face similar daily challenges. Future studies should explore how care configurations, staffing structures, and institutional routines influence the applicability of these design principles in long-term care settings and across diagnoses.

The racial and ethnic composition of our sample was predominantly White. Although some participants and therapists discussed culturally patterned IADLs—such as food traditions in Hispanic households—examining how cultural identity shapes meaningfulness, caregiving roles, and preferred activities was beyond the scope of this study. Similarly, while our findings touched on gendered patterns in how participants related to certain IADLs, gender was not a primary analytic focus. Future work would benefit from more culturally diverse samples and study designs intentionally structured to surface a broader range of cultural and gendered perspectives, including through community partnerships or international collaborations. This would help ensure that VR systems support varied cultural practices and avoid reinforcing Western-centric or normative defaults.

Finally, our findings reflect the perspectives of PwD, caregivers, and OTs who participated in our interview-based design probes. While these perspectives offered rich insights into everyday routines and design priorities, future work should investigate how these themes unfold through longitudinal VR use, multi-stakeholder co-design, and deployment studies that capture changing needs over time. Such efforts will help refine the ecological validity of IADL-based VR systems and ensure that they remain responsive, equitable, and feasible across diverse living and care contexts.

7 Conclusion

Our study examines how IADL-based VR can be designed to support PwD in their everyday lives, drawing on in-depth interviews with PwD, caregivers, and OTs. We define *realistic* tasks and virtual environments as those that align with real-world cognitive and functional demands, while *useful* VR is achieved when tasks evoke behavior that is reflective of how PwD act in real-life IADLs and meaningfully supports rehabilitation practice. Ecological validity is articulated as verisimilitude and face validity—as opposed to veridicality—as the most appropriate foundations for designing IADL-based VR. We further reframe IADL-based VR from a narrow training or assessment tool into a multi-functional platform for collaborative dementia care, enabling practice, safety assessment, adaptation, and caregiver and therapist involvement.

Our findings point to an opportunity for the HCI community to pivot research efforts toward IADL-based VR, guided by a refined understanding of ecological validity and design implications. By embedding VR in the lived realities of PwD and their caregiving ecosystems, future systems can better foster independence, strengthen relationships, and enhance quality of life.

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